

Optional task

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Domain : AI & ML

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```
...      age      sex      bmi  children smoker      region      charges
0      19  female  27.900           0      yes southwest  16884.92400
1      18   male  33.770           1      no  southeast  1725.55230
2      28   male  33.000           3      no  southeast  4449.46200
3      33   male  22.705           0      no northwest  21984.47061
4      32   male  28.880           0      no northwest  3866.85520

      age      sex      bmi  children smoker      region      charges
0 -1.438764 female -0.453320 -0.908614   yes southwest  0.298584
1 -1.509965   male  0.509621 -0.078767   no  southeast -0.953689
2 -0.797954   male  0.383307  1.580926   no  southeast -0.728675
3 -0.441948   male -1.305531 -0.908614   no northwest  0.719843
4 -0.513149   male -0.292556 -0.908614   no northwest -0.776802
```

[2]

✓ 0s

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
```

[3]

✓ Os

```
df = pd.read_csv("insurance.csv")
```

```
print(df.head())
```

▼

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
✓ ... (1338, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
None
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

[6]

✓ 0s



```
print("Duplicate Rows:", df.duplicated().sum())
```

```
df.drop_duplicates(inplace=True)
```



... Duplicate Rows: 1

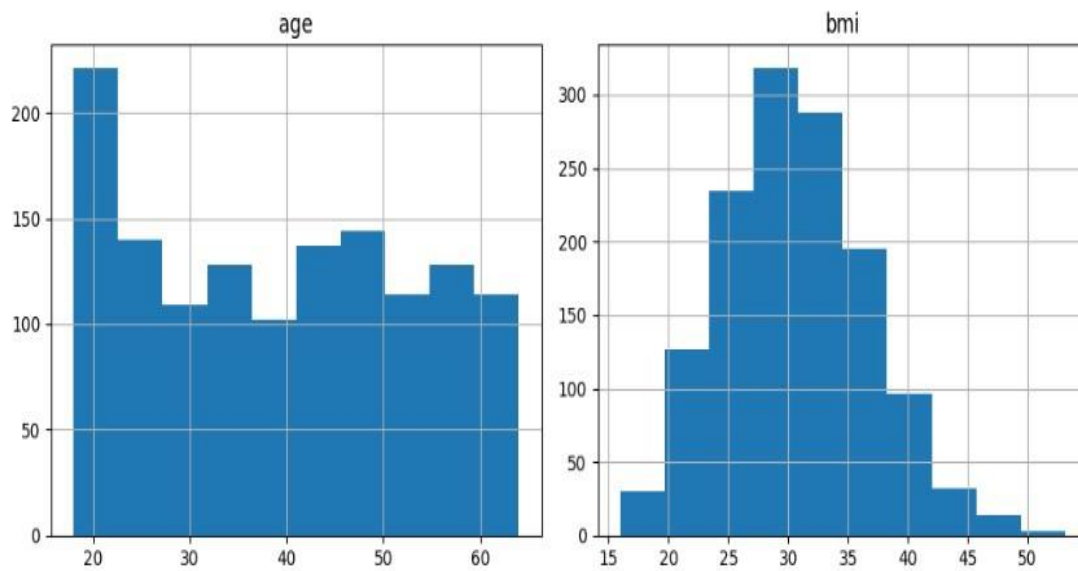
[7]

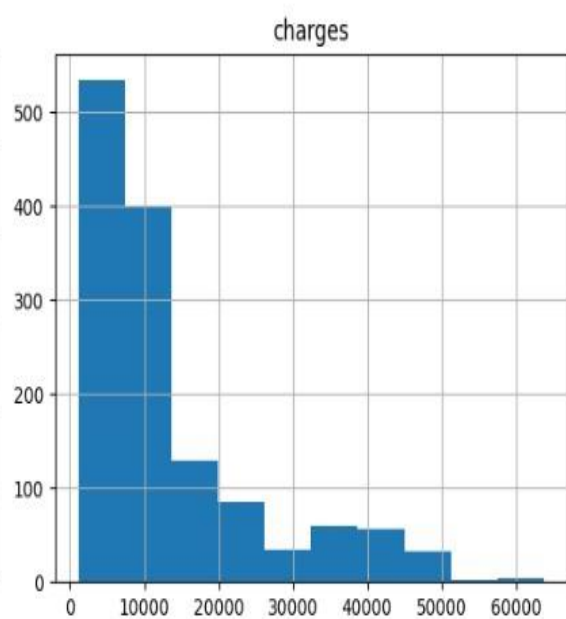
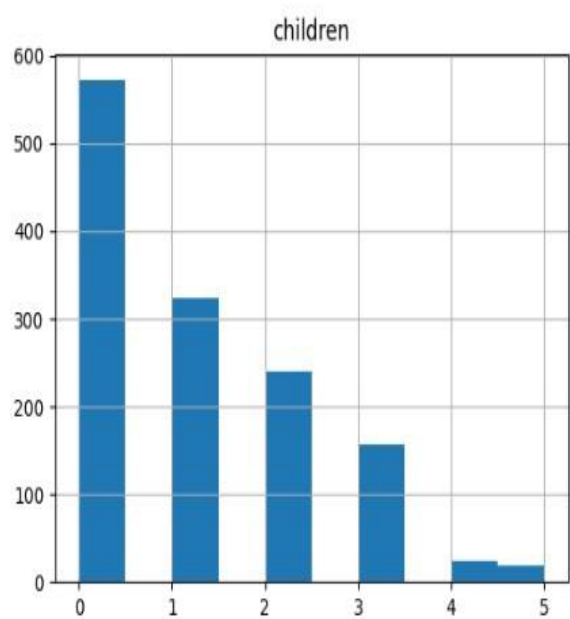
✓ 0s

```
df.hist(figsize=(10,8))  
plt.tight_layout()  
plt.show()
```

✓

...





[5]

✓ 0s

```
print(df.isnull().sum())
```

▼

```
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```


[8]
✓ Os



```
sns.countplot(x='sex', data=df)

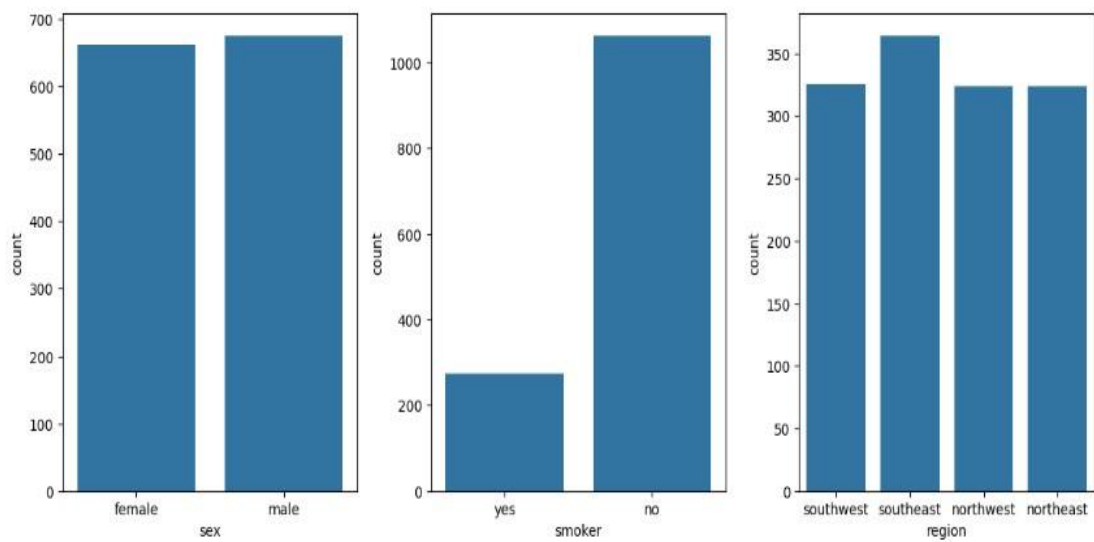
plt.subplot(1,3,2)
sns.countplot(x='smoker', data=df)

plt.subplot(1,3,3)
sns.countplot(x='region', data=df)

plt.tight_layout()
plt.show()
```



...



[9]
✓ 0s

```
▶ X = df.drop('charges', axis=1)

y = df['charges']

print(X.head())

print(y.head())
```

▼ ...

	age	sex	bmi	children	smoker	region
0	19	female	27.900	0	yes	southwest
1	18	male	33.770	1	no	southeast
2	28	male	33.000	3	no	southeast
3	33	male	22.705	0	no	northwest
4	32	male	28.880	0	no	northwest

0 16884.92400
1 1725.55230
2 4449.46200
3 21984.47061
4 3866.85520
Name: charges, dtype: float64

v

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520

✓	...	age	sex	bmi	children	smoker	region	charges
	0	-1.440418	0	-0.453160	-0.909234	1	3	0.297857
	1	-1.511647	1	0.509422	-0.079442	0	2	-0.954381
	2	-0.799350	1	0.383155	1.580143	0	2	-0.729373
	3	-0.443201	1	-1.305052	-0.909234	0	1	0.719104
	4	-0.514431	1	-0.292456	-0.909234	0	1	-0.777499

```

✓ ... <class 'pandas.core.frame.DataFrame'>
      Index: 1337 entries, 0 to 1337
      Data columns (total 7 columns):
      #   Column      Non-Null Count  Dtype
      ---  ---
      0   age         1337 non-null   float64
      1   sex         1337 non-null   int64
      2   bmi         1337 non-null   float64
      3   children    1337 non-null   float64
      4   smoker      1337 non-null   int64
      5   region      1337 non-null   int64
      6   charges     1337 non-null   float64
      dtypes: float64(4), int64(3)
      memory usage: 83.6 KB
      None

               age         sex         bmi         children         smoker \
count  1.337000e+03  1337.000000  1.337000e+03  1.337000e+03  1337.000000
mean   -6.643070e-17   0.504862 -2.285216e-16  3.852980e-17   0.204936
std     1.000374e+00   0.500163  1.000374e+00  1.000374e+00   0.403806
min    -1.511647e+00   0.000000 -2.411119e+00 -9.092342e-01   0.000000
25%    -8.705798e-01   0.000000 -7.171725e-01 -9.092342e-01   0.000000
50%    -1.582291e-02   1.000000 -4.320166e-02 -7.944162e-02   0.000000
75%     8.389340e-01   1.000000  6.619260e-01  7.503509e-01   0.000000
max     1.764921e+00   1.000000  3.684136e+00  3.239729e+00   1.000000

               region         charges
count  1337.000000  1.337000e+03
mean     1.516081 -1.009747e-16
std       1.105208  1.000374e+00

```